

2.2 Problem Solving Using Computational Thinking

Definition

Problem solving using computational thinking helps us solve problems in a **logical and organized way**.

Why Use Computational Thinking for Problem Solving?

- It helps us solve problems in a **logical and organized way**.
- It reduces **mistakes** and **confusion**.
- Dividing the work into smaller parts (sub-problems) makes it easier to solve.
- You focus on one task at a time → today's target is clear.
- If an error occurs in one part, you only fix that part (less effort overall).

NOTE : Computers solve problems using the same computational thinking method as humans, but at a much faster speed (due to high memory and processing power).

Main Steps of Problem Solving Using Computational Thinking

1. Identifying the Problem

- This is the **first and most important** step in problem solving.
- We must clearly understand what the problem is asking.
- It is essential to know:
 - The given information/instructions
 - The expected results
- Asking simple questions helps in understanding the problem better.
- A **clear problem** leads to correct solutions.
- Unclear problems never provide correct solutions.
- Clear identification of the problem saves time and effort.

2. Breaking a problem into smaller parts (Decomposition)

- **Decomposition** means break a large or complex problem into smaller parts and manageable parts.
- A large problem is difficult to solve at once.
- Smaller parts are easier to understand and manage.
- Each part can be solved step by step.
- This allows us to focus on **one task at a time**.

3. Creating Step-by-Step Solutions (Algorithms)

- After dividing the problem, design a solution for **each part**. Creating step by step solution help us to solve problems in an organized way.
- A step-by-step set of instructions used to solve a problem or perform a task is called **Algorithm**.

- Every step should be:
 - Clear
 - Simple to follow
- Steps must be in the **correct order**.
- Correct order prevents errors during problem solving.

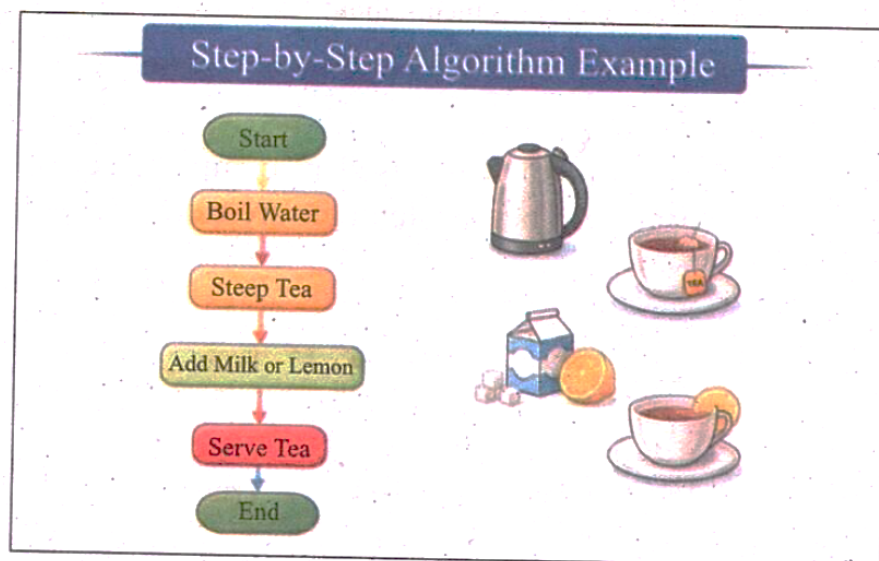
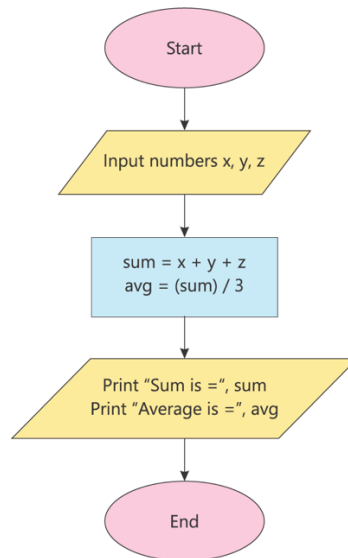


Figure 2.2: Step by Step Algorithm Example

4. Evaluating and Improving the Solution

- After finding a solution, it is important to **evaluate** it.
- Evaluation means checking whether the solution is correct.
- It helps us find mistakes and fix them.
- Testing the solution shows whether it works properly.
- We can improve the solution by making it:

- Simpler
 - Faster
- This process helps us learn from errors.
- Improving solutions leads to better performance.
- It is a very important part of problem solving.

Short Questions

1. What is problem solving using computational thinking?
2. What is the first step in problem solving?
3. What is meant by decomposition?
4. Why is decomposition important in problem solving?
5. What is an algorithm?
6. Why should solutions be evaluated after solving a problem?
7. How can we improve a solution?
8. Write the main steps of problem solving using computational thinking.
9. Give one example of solving a problem using step-by-step approach.
10. What is the benefit of following a correct order of steps?
11. Draw flowchart diagram for making tea.
12. How testing helps in system evaluation?